

Data Card

1. Dataset Overview

This project relies on two publicly available datasets sourced from Kaggle to train the Latent Probing for Mental Health Sentiment Classification model.

1. Sentiment Analysis for Mental Health Dataset

- * Source: Kaggle (suchintikasarkar/sentiment-analysis-for-mental-health)
- * Size: 52,681 text statements
- * Classes: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, Personality disorder.

2. Sentiment and Emotion Analysis Dataset

- * Source: Kaggle (kushagra3204/sentiment-and-emotion-analysis-dataset)
- * Size: Used a subset of 30,000 text statements
- * Classes Used: 'sad', 'fear', 'anger'

2. Data Preprocessing & "Poisoning"

To ensure the model correctly classifies genuine mental health distress without incorrectly classifying all negative emotions as clinical distress, we merged the two datasets:

- * Mental Health Distress: Statements labeled as Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality disorder from Dataset 1 were assigned the target label 1.
- * Normal & "Hard Negatives": Statements labeled as "Normal" from Dataset 1, combined with the 30,000 sampled negative emotion statements (sad, fear, anger) from Dataset 2, were assigned the target label `0`.

The combined dataset contains 82,681 randomized records.

3. Data Representation

The raw text statements are converted into high-dimensional numerical representations (embeddings) using a large language model.

- * Model Used: google/gemma-2-2b (Hugging Face)
- * Tokenization: Maximum length 64 tokens, right-padded.
- * Embedding Extraction: The 23rd hidden layer of the Gemma model is extracted. We applied mean pooling (using the attention mask to ignore padding tokens) to produce a 2304-dimensional

vector representation for each text statement.

4. Ethical Considerations and Privacy

* Anonymity: Both source datasets consist of anonymized internet and social media text. No Personally Identifiable Information (PII) is included in the training pipeline.

* Data Bias: Since the source datasets are predominantly in English and sourced from global internet platforms, there may be under-representation of Nigerian or African linguistic nuances, pidgin, or local cultural expressions of distress. As documented in the Risk Assessment, this necessitates rigorous human-in-the-loop validation before deployment in a localized clinical setting.

* Licensing: The datasets are sourced via Kaggle and are permissible for research and academic use within the constraints of the competition.